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Exam 4: NE533: Nuclear Fuel Performance

Show all work. Label question number in your response. Pay attention to units. Point values correspond to expected depth of response.

1. (18 pts) A ZIRLO cladding tube is in reactor at 625 K for 500 days. The initial wall thickness is 600 μm .
 - a) Estimate the oxide thickness after this time?
 - b) Assuming the hydrogen pickup fraction is 14%, what is the weight ppm of hydrogen in the cladding? Assume $\text{PBR} = 1.56$, $\rho_{\text{Zr}} = 6.5 \text{ g/cc}$, $\rho_{\text{ZrO}_2} = 5.68 \text{ g/cc}$.
2. (2 pts) What is the rate-limiting step in the aqueous corrosion of Zr cladding?
3. (8 pts) Where do hydrides form in the cladding? Why? What impacts do hydrides have?
4. (12 pts) What is a RIA? What is a typical RIA in a PWR or BWR. Describe what happens during a RIA.
5. (12 pts) What is a LOCA? Describe the impacts of a LOCA. How is it different than a RIA?
6. (4 pts) How does burnup impact the type of failure and the failure probability during an accident?
7. (6 pts) What are the four pathways to make the fuel/cladding system more accident tolerant?
8. (6 pts) List one near-term ATF concept and why it is of interest.
9. (6 pts) What happens to zirconium cladding when it is exposed to high temperature steam?
10. (6 pts) List three examples of limiting phenomena governing LWR operation.
11. (6 pts) What fuel performance and safety impacts does CRUD have?
12. (6 pts) List two water chemistry controls that have been implemented in LWRs, including why they were implemented.
13. (8 pts) List and describe two key microstructural phenomena that are unique to MOX fuels.

$$1) T = 625 K, t = 500 \text{ days} - t_0 = 600 \mu\text{m}$$

$$t^* = 6.62 \cdot 10^{-7} \exp\left(\frac{11944}{T}\right) = 133.007 \text{ days}$$

$$\delta = \delta^* + K_L(t - t^*) \quad \delta^* = 5.1 \exp\left(-\frac{550}{T}\right) = 2.11539 \mu\text{m}$$

$$K_L = 7.48 \cdot 10^{-6} \exp\left(-\frac{12500}{T}\right) = 0.015417 \frac{\mu\text{m}}{\text{d}}$$

$$a) \delta = 7.77348 \mu\text{m}$$

$$b) f = 0.14$$

$$PBR = 1.56$$

$$C_H^{\text{clad}} = \frac{2f \cdot \delta \cdot \rho_{\text{oxide}} \cdot f_{\text{ZrO}_2}^0 \cdot \frac{M_H}{M_O}}{(t - \frac{\delta}{PBR}) \cdot \rho_{\text{metal}}} \cdot 10^6$$

$$\rho_{\text{metal}} = 6.5 \text{ g/cc}$$

$$\rho_{\text{oxide}} = 5.68 \text{ g/cc}$$

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$$t = 600 \mu\text{m}$$

$$\delta = 7.77348 \mu\text{m}$$

$$f_{\text{ZrO}_2}^0 = \frac{16 \cdot 2}{16 \cdot 2 + 91} = 0.260163$$

$$\frac{M_H}{M_O} = \frac{1}{16}$$

$$C_H^{\text{clad}} = \frac{2 \cdot 0.14 \cdot 7.77348 \cdot 5.68 \cdot 0.260163 \cdot \frac{1}{16}}{(600 - \frac{7.77348}{1.56}) \cdot 6.5} \cdot 10^6$$

calculator error?

$$= 9.15073 \text{ wt. ppm}$$

2) Diffusion of oxygen through the oxide layer

3) Hydrides form towards the outside of the cladding due to lower temperatures as well as ^{high stress} areas such as crack tips due to increased potential there. Hydrides are very brittle and thus significantly increase probability of failure/cracking.

4) Reactivity Insertion Accident. In a PWR, this is typically a Control Rod Ejection, and in a BWR, this is typically a control rod dropping out of the core. In an RIA, power spikes very quickly, causing a massive temperature spike, causing the fuel to expand and can lead to DNR.

4 (continued) DNB is more likely to occur in lower burnup fuel, while PCMI is more likely in high burnup fuel. An increase in fuel rod pressure also occurs, which can lead to ballooning & burst of the cladding.

^{12/12} 5) Loss of Cooling Accidents occur when there is a loss of cooling in the core, typically due to a pipe burst in the primary system. This leads to elevated fuel temperatures, causing more oxidation. Since oxidation of the cladding is exothermic, this leads to even more oxidation leading to production of hydrogen, which can form hydrides and is very hazardous as a gas. (flammable, explosive)
When ECCS (emergency core cooling system) is activated, the thermal shock can cause cladding fracture & failure. Ballooning & burst of the cladding can also occur in LOCAs due to increased temperatures, fuel expansion, ideal gas law, etc. LOCAs occur on a longer timescale (minutes) than RAs (less than a second).

^{4/4} 6) Burnup can make PCMI more likely than DNB, due to the closing of the gap. Burnup generally increases failure probability, as cladding becomes more brittle, the gap shrinks/disappears, conductivity degrades, due to fission products, etc.

^{5/6} 7) - Increase time to melt - kind of
- Increase time to cladding failure - kind of
- Decrease fission product release ✓
- Decrease oxidation of cladding ✓

8) Ceramic coating of cladding has potential to greatly reduce cladding oxidation by protecting the Zr from oxidation. $\frac{6}{6}$

9) Zirconium cladding oxidizes $\frac{5}{6}$ when exposed to high T steam. If the steam is high enough temperature, it can undergo a phase change from α to β .
- needed more

10) Cladding elongation / bow, cladding oxidation & hydride pickup, $\frac{6}{6}$ power to melt.

11) CRUD can decrease thermal conductivity, become radioactive and flake off onto non-core structures, and boron pickup in CRUD causes a $\frac{6}{4}$ CRUD-induced power shift.

12) Boron and lithium - Boron ^{soluble} used in PWRs to control reactivity, lithium to balance the pH bulk to neutral.

$\frac{6}{4}$ 2mc - Used to decrease the induced radioactivity of the assemblies, by taking the place of other nuclides like Co-60.

13) Central void - Fission ^{voids} products migrate up thermal gradient in the fuel to form a central void columnar region - Grains grow to form a columnar structure near the central void region.

$\frac{3}{8}$ Additionally, oxygen will migrate towards lower temp regions of the fuel, causing the center of the pellet to have a higher concentration of fissile material, while the outer region of the pellet has more oxygen content.

